
DATA1010 - INTRODUCTION TO DATA SCIENCE

SPRING 2026

FINAL PROJECT SUBMISSION GUIDELINE

SUBMISSION AND DEADLINE

Each group must prepare **four items**:

1. One written report (PDF)
2. One program package (Python notebook, `.ipynb` / GitHub repository)
3. One presentation slide (PDF or PowerPoint)
4. One poster (PDF)

Deadline: all items (written report, code package, slides, and poster) must be submitted to Canvas **by 23:59 on Monday, 1 June 2026**.

WRITTEN REPORT

- **Format.** The report must be in PDF format, with maximum **6 pages** (including references and contribution table). The report must be readable when printed (recommended: A4 size, standard margins, 11–12 pt font size).
- **Content.** The report must contain clear citations and references for any external sources used, including generative AI. Any graphs or figures must contain proper titles, labels, and descriptions.
- **Structure.** The report should follow the sample given in Appendix A. In particular, it must contain the **Contribution declaration** table.

PROGRAM PACKAGE

- **Format (choose one option).**
 - **Option A: Python notebook (`.ipynb`).**
 - * Upload one copy to Canvas.
 - * Upload one copy to Google Colab and insert the Colab link into the written report.
 - * Ensure the link is accessible to instructors until your final grades are released.
 - **Option B: GitHub repository.**
 - * Upload one copy (`.zip`) to Canvas.
 - * Insert the repo link into the written report.
 - * Ensure the link is accessible to instructors until your final grades are released.
- **Content.** The code must include all computations, experiments, and results reported in the written report.
- **Structure.** Organize the code into clearly labeled sections corresponding to each question.

PRESENTATION SLIDE

- **Format.** The slide must be in either PDF or PowerPoint format.
- **Content.** Focus on data collecting+processing, methodology, and experiment results.
- **Structure.** Divide the presentation into sections by question.

POSTER

- **Format.** The poster must be in 1 page A0, portrait, and PDF format.
- **Content.** Summarize the dataset, methodology, and key results.

OTHER NOTES

It is strongly recommended to begin the project as early as possible. For any questions or clarifications, contact the instructor **by 23:59 on Thursday, 28 May 2026**.

UPDATE

Any changes will be posted here.

APPENDIX A: QUESTION SHEET

In this final project, each group will collect a practical, real-world dataset and analyze it using both supervised learning and unsupervised learning techniques.

1. Task 1: Data Collection and Data Processing

Each group must collect a real dataset (self-collected; **no online public datasets allowed**). The dataset must meet the following criteria:

- 01 binary target variable (to be used in supervised learning).
- At least 04 additional features (to be used in both supervised and unsupervised learning).
- At least 50 valid training samples and another 50 valid testing samples **after** data cleaning.
- The cleaned dataset must be split equally into a training set and a testing set.

Your written report must include:

- Data collection methods: Describe how the data was collected (e.g., in-person or online form, interview, duration, population/sampling method, etc.)
- Data privacy considerations: Explain how you handled sensitive information and ensured participant privacy and data security
- Data processing: Describe how you handled missing or erroneous values, normalization, feature encoding, and any preprocessing techniques used
- Data visualization: Provide visualizations that describe feature distributions, relationships, and correlations (e.g., histograms, scatter plots, correlation heatmaps)

2. Task 2: Supervised learning

Use the collected dataset to predict the binary target variable using two classification models.

Required steps and content for the report:

- Research question: what are you trying to predict/analyze/control?
- Model implementation: Apply both Logistic Regression and Support Vector Machine (SVM)
- Model evaluation: Report and compare performance on the training and testing sets using appropriate metrics (e.g., accuracy, precision, loss/error values)
- Analysis: Interpret and explain your results. Discuss which model performed better and why.

3. Task 3: Unsupervised learning

Explore patterns and groupings within your dataset using clustering techniques.

Required steps and content for the report:

- Research question: what are you trying to predict/analyze/control?
- Model implementation: Apply K-Means clustering on the dataset (excluding the binary target variable)
- Evaluation: Report clustering results and relevant metrics (e.g., loss, number of clusters)
- Analysis: Visualize and interpret the clusters. Explain the choice of number of cluster k . Explain if the clusters align with any known patterns or segments in your data.

APPENDIX B: WRITTEN REPORT SAMPLE

DATA1010 Final Project: Group

Member list:

No.	Name	Student ID	Email
1	Jane Doe	V202310000	Jane.d@email
2	John Doe	V202310001	John.d@email
...			

Link to Google Colab: [Insert link here]

1. Introduction, Data Collection and Data Processing

In this project, we aim to study students' habits based on their physical and lifestyle features...

We collected data in person using ..., and data was collected from ... to ...

The questionnaires included 6 questions: Q1: ..., Q2: ..., etc.

The target binary variable is **Gender** (0/1), features are **Age** (years old), **Height** (cm), ...

Summary of data processing: ...

Visualization: ...

2. Supervised Learning

We apply supervised learning models: Logistic Regression and Support Vector Machine (SVM) to predict gender.

These two models are described as follows...

We report the accuracy of these models as follows...

We observed that SVM slightly outperformed logistic regression. One of the reasons is ...

3. Unsupervised Learning

We applied K-Means clustering to cluster students. We choose $k = 2$ because ...

We report the numerical performance and accuracy as follows...

The clusters mostly separated male and female participants, however...

4. References

5. Contribution declaration

Task	Part	Sub-task	Tom	Rose	Kate	Bob
1	Write report	Design questionnaires	✓✓	✓	✓	
1	Write report	Clean data	✓		✓	
...						
1	Write code	Visualize histogram				✓
1	Write code	Convert data				✓
...						
1	Prepare slide	Prepare slide		✓	✓	
2	...	...	...	...	...	...

Note: Use ✓ to indicate contributors, use ✓✓ to indicate student in charge (who will answer Q&A about that sub-task). Each sub-task (row) selects at most 3 student, and each student is in charge at at least 1 sub-task.

APPENDIX C: GRADE RUBRIC OF WRITTEN REPORT AND PROGRAM PACKAGE

Content	Full	Level of Performance		
		Unsatisfactory	Satisfactory	Excellent
(criteria)		Missing required sections, unclear writing/structure, missing citations/references, or weak/incorrect interpretation of results.	Complete structure with clear writing and basic explanations; citations and figures mostly correct; minor issues may be acceptable.	Complete, well-organized, and clearly written; strong interpretation/discussion; figures and citations are thorough and professional. MUST include extra exploration, analysis and optimization instead of only a single accuracy or loss value.
Written Report				
Task 1	30	0/5/10/15	20/25	30
Task 2	30	0/5/10/15	20/25	30
Task 3	30	0/5/10/15	20/25	30
(criteria)		Code is missing/incorrect, does not run end-to-end, or results are not reproducible/traceable to the report.	Runs and reproduces reported results; code is reasonably organized into sections with basic comments/outputs.	Clean, well-structured, and reproducible; clear sectioning and documentation; includes sanity checks and thoughtful extra exploration where appropriate.
Program Package				
Code	10	0/2/4/6	8	10
All	100	≤ 50	≤ 83	≤ 100

* Note:

- The above rubric applies to the entire group. All group members will typically receive the same grade. However, individual member(s) who demonstrate significant or exceptional contributions may be considered for bonus points at the instructor's discretion. Bonus points, ranging from 1 to 5 (on a 100-point scale), are not guaranteed and may involve additional interview questions. They are awarded selectively and not necessarily granted to every group or every project.
- Any late/missing submissions without justification will result in 0 points for the entire group.
- Students who do not contribute to the project or are irresponsible for assigned tasks are subject to deductions, up to –100. This includes not coming to class/lab to work with group members, not being reachable during the project period, not accepting any task, or not completing tasks that appear in the attached contribution table under their name.

APPENDIX D: GRADE RUBRIC OF PRESENTATION DAY

Content	Full	Level of Performance		
		Unsatisfactory	Satisfactory	Excellent
Materials				
(criteria)		Missing required components; unclear/incorrect content; poor readability/organization; figures/tables are unlabeled or misleading.	Covers required content with mostly correct results and clear organization; visuals are readable with minor issues.	Polished and professional; clear and accurate narrative; strong visuals; insightful interpretation and takeaways.
Poster	25	0/5/10/15	18/20	25
Slides	25	0/5/10/15	18/20	25
Oral Presentation and Interview				
(criteria)		Unclear delivery; poor time management; cannot explain methods/results; major issues answering questions.	Mostly clear delivery and pacing; explains core ideas and results; answers most questions adequately.	Confident, well-paced delivery; strong command of methods/results; engages audience; answers questions thoughtfully and accurately.
Oral presentation	25	0/5/10/15	18/20	25
Interview / Q&A	25	0/5/10/15	18/20	25
All	100	≤ 60	≤ 80	≤ 100

* Note:

- The above rubric applies to the entire group. All group members will typically receive the same grade for poster/slides and the oral presentation.
- Interview/Q&A questions may be directed to individual members. If a member cannot answer questions about the project adequately, individual deductions may be applied at the instructor's discretion.